

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286235

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

ZHAO; Hengxin et al.

BATTERY MODULE AND BATTERY PACK

Abstract

Provided are a battery module and a battery pack. The battery module includes multiple battery cells, multiple first connection members and multiple second connection members. A battery cell has an end surface, and the end surface is provided with a tab. The multiple first connection members and the multiple second connection members are flexible members. A first connection member has a first connection portion, a second connection portion and a bend. Two ends of the bend are connected to the first connection portion and the second connection portion, separately. The multiple battery cells are disposed in a superimposed manner. The first connection portion is connected to tabs of two adjacent battery cells. The second connection member is connected between second connection portions of two adjacent first connection members, the first connection portion and the second connection portion are superimposed on the end surface.

Inventors: ZHAO; Hengxin (Huizhou, CN), LI; Zhimiao (Huizhou, CN), HONG; Guoqiang (Huizhou, CN), GUAN; Xinli (Huizhou, CN), HUANG; Guojian (Huizhou, CN), QI; Dongrong (Huizhou, CN), DOU; Peng (Huizhou, CN), LIU; Pengcheng (Huizhou, CN)

Applicant: EVE ENERGY CO., LTD. (Huizhou, CN)

Family ID: 93930924

Appl. No.: 18/946267

Filed: November 13, 2024

Foreign Application Priority Data

CN	202420422777.6
WO	PCT/CN2024/108689

Mar. 05, 2024
Jul. 31, 2024

Publication Classification

Int. Cl.: H01M50/521 (20210101); H01M50/211 (20210101); H01M50/291 (20210101)

U.S. Cl.:

CPC H01M50/521 (20210101); H01M50/211 (20210101); H01M50/291 (20210101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application claims priority to Chinese Patent Application No. 202420422777.6, filed on Mar. 5, 2024, and PCT application PCT/CN2024/108689 filed on Jul. 31, 2024, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The present application relates to the technical field of lithium batteries, for example, a battery module and a battery pack.

BACKGROUND

[0003] The pouch lithium-ion battery cell has the “deep breathing” effect in the free state during the charging and discharging process, which specifically presents as a significant volume change in the battery cell. Compared with the no power state, a lithium metal battery expands in the full power state, that is to say, the “deep breathing” effect of the battery cell causes expansion of the battery cell, which causes the non-uniformity in thickness, the slip phenomenon between battery cells, and the deformation of the lithium metal battery, and thus is disadvantageous for the charging and discharging cycle. Therefore, the restraining force needs to be applied in the thickness direction of the battery cell to ensure the cycle.

[0004] To improve the energy density of the battery, multiple battery cells are disposed within the battery pack, and the multiple battery cells are connected in series and in parallel through a printed circuit board (PCB). To prevent the PCB and tabs connected to the PCB from being broken, restraints are generally added to four side surfaces of the battery cell. However, the slip of the battery cell due to the “deep breathing” effect may cause the dislocation between the battery cell and the PCB, and may, in severe cases, cause the PCB to break or cause the tabs to be torn apart, resulting in the overall failure of the battery pack and the low reliability and poor safety of the battery pack.

SUMMARY

[0005] The present application provides a battery module. The battery module includes multiple battery cells, multiple first connection members and multiple second connection members. A battery cell includes an end surface, and the end surface is provided with a tab. The multiple first connection members and the multiple second connection members are flexible members, a first connection member includes a first connection portion, a second connection portion and a bend, the first connection portion and the second connection portion are in sheet form, and two ends of the bend are connected to the first connection portion and the second connection portion, respectively. The multiple battery cells are disposed in a superimposed manner, the first connection portion is connected to tabs of two adjacent battery cells, the multiple second connection members are in sheet-like form, a second connection member is configured to connect second connection portions of two adjacent first connection members, and the second connection member, the first connection portion and the second connection portion are superimposed on the end surface of the battery cell.

[0006] The present application provides a battery pack. The battery pack includes a box body and a battery module. The battery module is disposed within the box body, and a side surface of the battery module is attached to an inner wall of the box body.

[0007] The beneficial effects of the present application are as follows. The multiple first connection members and the multiple second connection members are provided so that the battery cells are separately connected through the first connection members, and the multiple first connection members and the multiple second connection members cooperate to replace the original integral PCB. When a battery cell expands, the first connection member and the second connection member can more easily absorb the expansion of the battery cell by generating deformation, preventing the tab from being torn off, and preventing the first connection member and the second connection member from being separated from the tab of the battery cell. In this manner, the stability of the connection between the battery cells can be ensured and the open circuit in the battery module is prevented. The second connection member, the first connection portion and the second connection portion are set to be superimposed on the end surface of the battery cell so that the overall shape of the battery module can be approximately a cuboid. Moreover, the first connection member and the second connection member have flexibility, so the external restraint structural member can provide compaction and restraint for six side surfaces of the battery module, the slip phenomenon in an expansion process of the battery cell is alleviated, the deformation of the battery module is reduced, and thus the reliability and safety of the battery module are improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic diagram one of a connection between a battery cell and a first connection member according to some embodiments of the present application.

[0009] FIG. 2 is a schematic diagram two of a connection between a battery cell and a first connection member according to some embodiments of the present application.

[0010] FIG. 3 is a schematic diagram three of a connection between a battery cell and a first connection member according to some embodiments of the present application.

[0011] FIG. 4 is a schematic diagram one of a connection between a battery cell, a first connection member and a second connection member according to some embodiments of the present application.

[0012] FIG. 5 is a schematic diagram two of a connection between a battery cell, a first connection member and a second connection member according to some embodiments of the present application.

[0013] FIG. 6 is a schematic diagram of a battery module according to some embodiments of the present application.

[0014] FIG. 7 is a schematic diagram one of a connection between a battery cell and a first connection member according to some other embodiments of the present application.

[0015] FIG. 8 is a schematic diagram two of a connection between a battery cell and a first connection member according to some other embodiments of the present application.

[0016] FIG. 9 is a schematic diagram three of a connection between a battery cell and a first connection member according to some other embodiments of the present application.

[0017] FIG. 10 is a schematic diagram of a connection between a battery cell, a first connection member and a second connection member according to some other embodiments of the present application.

[0018] FIG. 11 is a schematic diagram of a battery module according to some other embodiments of the present application.

[0019] FIG. 12 is a schematic sectional view of a battery pack according to some embodiments of the present application.

[0020] FIG. 13 is an exploded schematic view of a battery pack according to some embodiments of the present application.

[0021] FIG. **14** is a schematic diagram of a battery pack according to some embodiments of the present application (one splice plate is not shown).

[0022] FIG. **15** is a schematic diagram of a second buffer member according to some embodiments of the present application.

[0023] FIG. **16** is a schematic diagram of a box body and a separator according to some embodiments of the present application (one splice plate is not shown).

[0024] FIG. **17** is a structural diagram of an electrode tab according to some embodiments of the present application.

LIST OF REFERENCE NUMBERS

[0025] **1** box body [0026] **11** splice plate [0027] **12** separator [0028] **13** current output seat [0029] **2** second buffer member [0030] **21** wiring groove [0031] **210** wire [0032] **22** weight reduction hole [0033] **3** battery module [0034] **31** power transmission side surface [0035] **32** battery cell [0036] **321** edge sealing portion [0037] **322** tab [0038] **3221** third connection portion [0039] **3222** fourth connection portion [0040] **323** body portion [0041] **33** first connection member [0042] **331** first connection portion [0043] **332** second connection portion [0044] **333** bend [0045] **34** second connection member [0046] **35** supporting member [0047] **36** first buffer member

DETAILED DESCRIPTION

[0048] In the description of the present application, terms “joined”, “connected” and “fixed” are to be understood in a broad sense unless otherwise expressly specified and limited. For example, the term “connected” may refer to “securely connected”, “detachably connected” or “integrated”, may refer to “mechanically connected” or “electrically connected”, or may refer to “connected directly”, “connected indirectly through an intermediary” or “internal connection between two elements or interaction between two elements”. For those of ordinary skill in the art, specific meanings of the preceding terms in the present application may be understood based on specific situations.

[0049] As shown in FIG. **6** and FIG. **11** (reference is made to FIG. **1** to FIG. **5**, and FIG. **7** to FIG. **10** for some of the reference numerals), the present application provides a battery module **3**. The battery module **3** includes multiple battery cells **32**, multiple first connection members **33** and multiple second connection members **34**. A battery cell **32** has an end surface, and the end surface is provided with a tab **322**. The multiple first connection members **33** and the multiple second connection members **34** are flexible members, that is, the multiple first connection members **33** and the multiple second connection members **34** have flexibility. The multiple first connection members **33** and the multiple second connection members **34** may be deformed under pressure. A first connection member **33** has a first connection portion **331**, a second connection portion **332** and a bend **333**. The first connection portion **331** and the second connection portion **332** are in sheet-like form. Two ends of the bend **333** are connected to the first connection portion **331** and the second connection portion **332**, respectively. The multiple battery cells **32** are disposed in a superimposed manner. The first connection portion **331** is connected to tabs **322** of two adjacent battery cells **32**. The multiple second connection members **34** are in sheet-like form, and the second connection member **34** is connected between second connection portions **332** of two adjacent first connection members **33**. The second connection member **34**, the first connection portion **331** and the second connection portion **332** are superimposed on the end surface of the battery cell.

[0050] The multiple first connection members **33** and the multiple second connection members **34** are provided so that battery cells **32** among the multiple battery cells **32** are separately connected through the first connection portion **331** and the second connection portions **332** of the first connection members **33** are connected through the second connection members **34** to replace an integral PCB. In this embodiment, when a battery cell **32** expands, the first connection member **33** and the second connection member **34** can more easily absorb the expansion of the battery cell **32** by generating deformation, to prevent the tab **322** from being torn off, and also prevent the first connection member **33** and the second connection member **34** from being separated from the tab **322** of the battery cell **32**, thereby ensuring the stability of connection between the battery cells **32**

and preventing the open-circuit in the battery module **3**. The second connection member **34**, the first connection portion **331** and the second connection portion **332** are disposed to be superimposed on the end surface of the battery cell **32** so that the overall shape of the battery module **3** may be approximately a cuboid. Moreover, the first connection member **33** and the second connection member **34** have flexibility, the external restraint structural member may provide compaction and restraint for six side surfaces of the battery module **3**, so that the slip phenomenon in an expansion process of the battery cell **32** is alleviated, the deformation of the battery module **3** is reduced, and thus the reliability and safety of the battery module are improved. [0051] In this embodiment, each of the first connection member **33** and the second connection member **34** is a nickel piece, and the first connection member **33** is bent to form the first connection portion **331**, the bend **333** and the second connection portion **332**.

[0052] In this embodiment, the first connection member **33** and the second connection member **34** made of the nickel piece are adopted so that the weight of the battery module **3** can be reduced to achieve a light-weight design.

[0053] With reference to FIGS. **6** and **11**, the battery module **3** includes a supporting member **35**. The battery cell **32** has a body portion **323** and an edge sealing portion **321**. The thickness of the body portion **323** is greater than the thickness of the edge sealing portion **321**. The supporting member **35** is disposed between edge sealing portions **321** of two adjacent battery cells **32**, the supporting member **35** abuts against a side surface of the edge sealing portion **321**, and side surfaces of a part of supporting members **35** are attached to both the first connection member **33** and the second connection member **34**, and side surfaces of the other part of supporting members **35** are separately attached to the second connection member **34**. In this embodiment, the battery cell **32** is a pouch battery, and the edge sealing portion **321** is an aluminum plastic film edge sealing, which belongs to an inherent structure of the pouch battery. The supporting member **35** is provided so that the supporting member **35** may fill a gap between edge sealing portions **321** of two adjacent battery cells **32**, thereby reducing the deformation of the edge sealing portion **321**. Moreover, the supporting member **35** may provide support for the first connection member **33** and the second connection member **34**, thereby reducing the curved degree of the first connection portion **331**, the second connection portion **332** and the second connection member **34**, and thus improving the reliability of the battery module **3**.

[0054] In this embodiment, the supporting member **35** is a flexible supporting block. It is to be understood that, the battery cell **32** may expand in a charging and discharging process, the volume of the battery cell **32** changes, and the supporting member **35** is configured as the flexible supporting block to have flexibility; therefore, the supporting member **35** may absorb the deformation of the battery cell **32** to some extent, thereby reducing the deformation of the battery module **3** and improving the reliability and the safety of the battery module **3**. In some embodiments, the supporting member **35** may be an ethylene-vinyl acetate copolymer (EVA copolymer) foam.

[0055] Referring to FIG. **6** and FIG. **11**, the hardness of supporting members **35** located on two sides of the battery module **3** may be greater than the hardness of supporting members **35** located in the center of the battery module **3**. The supporting members **35** located on the two sides of the battery module **3** are directly in contact with the external restraint structural member to limit the battery cells **32**.

[0056] A first buffer member **36** is disposed between two adjacent battery cells **32**, and the first buffer member **36** is in sheet-like form. The first buffer member **36** is provided so that the deformation of the battery cell **32** may be absorbed, the deformation of the battery module **3** may be reduced, and the reliability and safety of the battery module **3** may be improved. The first buffer member **36** may be an EVA foam.

[0057] As shown in FIG. **17**, the tab **322** of the battery cell **32** includes a third connection portion **3221** and a fourth connection portion **3222**. The fourth connection portion **3222** of the tab **322** is

connected to the core in the battery cell **32** through the third connection portion **3221** of the tab **322**. Both the third connection portion **3221** and the fourth connection portion **3222** are in sheet-like form. The fourth connection portion **3222** is perpendicularly connected to the third connection portion **3221**, that is, the sheet-like tab **322** is bent 90° to form the third connection portion **3221** and the fourth connection portion **3222**. The fourth connection portion **3222** is parallel to the end surface of the battery cell **32**, and the fourth connection portion **3222** is connected to the first connection portion **331**. The fourth connection portion **3222** is disposed to be parallel to the end surface of the battery cell **32** and be connected to the first connection portion **331** so that bending of the connection position between the tab **322** and the first connection member **33** can be avoided, thereby reducing the difficulty of connecting the tab **322** and the first connection member **33** and reducing the impact of bending on the connection effect, and thus improving the reliability of connecting the tab **322** and the first connection member **33**.

[0058] In this embodiment, the first connection portion **331** is welded to the tab **322**, and the second connection portion **332** is also welded to the second connection member **34**. An ultrasonic welding manner, a laser welding manner or other welding manners may be used.

[0059] The battery cell **32** generally has two tabs **322**. One tab **322** is used as a positive electrode, and the other tab **322** is used as a negative electrode. Two tabs **322** of a battery cell **32** shown in FIG. **1** to FIG. **6** are located on the same side of the battery cell **32**, which is denoted as a single-head-tab battery. The electrode tab **322** of the single-head electrode-tab-outlet battery projects from the same end surface of the battery cell **32**. Two tabs **322** of a battery cell **32** shown in FIG. **7** to FIG. **11** are located on two sides of the battery cell **32**, respectively, which is denoted as a dual-head-tab battery. The tabs **322** of the dual-head tab-outlet battery project from the two end surfaces of the battery cell **32**.

[0060] Referring to FIG. **1** to FIG. **6**, for the single-head tab-outlet battery, the battery module **3** may be manufactured by using the following steps.

[0061] In step **101**, referring to FIG. **1**, a (positive) tab **322** of one battery cell **32** (single-head electrode-tab-outlet battery) and a (negative) tab **322** of another battery cell **32** are connected and combined by using the first connection member **33** (nickel sheet) through ultrasonic welding.

[0062] In step **102**, referring to FIG. **2**, on the basis of FIG. **1**, tabs **322** of two battery cells **32** (single-end-tab batteries) are bent by 90° , the two battery cells **32** (single-end-tab batteries) are superimposed, one first buffer member **36** is placed between the two battery cells **32** (single-end-tab batteries), and then the first connection member **33** (a nickel sheet) is bent by 90° so that the first connection member **33** has an L-shape. A battery combination is formed in this step.

[0063] In step **103**, referring to FIG. **3**, multiple battery combinations shown in FIG. **2** are superimposed, and it is ensured that one first buffer member **36** is placed between every two battery cells **32** (single-end-tab batteries).

[0064] In step **104**, referring to FIG. **4**, on the basis of FIG. **3**, the battery combinations shown in FIG. **2** are connected by using the second connection members **34** according to the series and parallel connection design requirements of circuits in the battery module **3**, and the second connection member **34** (nickel sheet) is connected between the second connection portions **332** of the first connection member **33** (nickel sheet) in an ultrasonic welding manner.

[0065] In step **105**, referring to FIG. **5**, on the basis of FIG. **4**, an excess portion of the first connection member **33** (nickel sheet) is cut off, and then the first connection member **33** (nickel sheet) continues to be bent by 90° to form the bend **333** of the first connection member **33**. In this case, the first connection portion **331** and the second connection portion **332** of the first connection member **33** are superimposed.

[0066] In step **106**, referring to FIG. **6**, on the basis of FIG. **5**, the supporting member **35** is filled into the gap between the battery cells **32**.

[0067] Referring to FIG. **7** to FIG. **11**, for the dual-head-tab battery, the battery module **3** may be manufactured by using the following steps.

[0068] In step **201**, referring to FIG. **7**, a (positive) tab **322** and a (negative) tab **322** of the battery cell **32** (dual-head-tab battery) are ultrasonically welded to two first connection members **33** (nickel sheet), respectively.

[0069] In step **202**, referring to FIG. **8**, the tab **322** is bent by 90° on the basis of FIG. **7**.

[0070] In step **203**, referring to FIG. **9**, the first connection member **33** (nickel sheet) is bent by 90° on the basis of FIG. **8**.

[0071] In step **204**, referring to FIG. **10**, multiple combinations of the battery cell **32** (dual-head-tab battery) and the first connection member **33** shown in FIG. **9** are stacked and superimposed, one first buffer member **36** is placed between every two battery cells **32** (dual-head-tab batteries), a groove of the tab **322** of the battery cell **32** is padded to the ultrasonic welding position of the tab **322**, and then the multiple battery cells **32** (dual-head-tab batteries) are connected by using the second connection members **34** (nickel sheet) according to the series and parallel connection design requirements of circuits in the battery module **3**.

[0072] In step **205**, referring to FIG. **11**, the first connection member **33** (nickel sheet) is bent by 90° on the basis of FIG. **10**. In this case, the first connection portion **331** and the second connection portion **332** of the first connection member **33** are superposed.

[0073] Referring to FIG. **12** to FIG. **16**, this embodiment further provides a battery pack. The battery pack includes a box body **1** and the battery module **3** in any one of the embodiments. The battery module **3** is disposed within the box body **1**, and a side surface of the battery module **3** is attached to an inner wall of the box body **1**. The box body **1** in this embodiment is denoted as the restraint structural member mentioned in the embodiments. The shape of the battery module **3** is a cube to reduce, the design difficulty of the restraint structural member, that is, the battery module **3** may be restrained by using an intra-box plane structure, thereby reducing the design difficulty of the battery pack and the cost.

[0074] The battery pack further includes a wire **210** and a second buffer member **2**. A current output seat **13** is disposed on the box body **1**. The battery module **3** has a power transmission side surface **31** (end surfaces of multiple battery cells **32** are spliced to form the power transmission side surface **31**). The second connection members **34** of the battery module **3** are disposed on the power transmission side surface **31**. Two ends of the wire **210** are connected to the current output seat **13** and the second connection members **34**, respectively. The second buffer member **2** is disposed between the power transmission side surface **31** and the inner wall of the box body **1**. The second buffer member **2** is provided so that the second buffer member **2** can absorb the expansion of the battery cell **32**, to reduce the extrusion of the wire and improve the reliability of the battery pack. The second buffer member **2** may be an EVA foam.

[0075] A wiring groove **21** is disposed within the second buffer member **2**, and the wire **210** is disposed within the wiring groove **21**. The wiring groove **21** is provided so that the extrusion of the battery cell **32** on the wire **210** in a collision state can be further reduced, and thus the reliability of the battery pack is improved.

[0076] Referring to FIG. **15**, the second buffer member **2** is also provided with a weight reduction hole **22**. The weight reduction hole **22** is provided so that the weight of the second buffer member **2** can be reduced to reduce the overall weight of the battery pack and implement the light-weight design of the battery pack.

[0077] Multiple battery modules **3** are disposed within the box body **1**. Referring to FIG. **16**, multiple separators **12** are also disposed within the box body **1**, the separator **12** is fixed to the inner wall of the box body **1**, a separator **12** is disposed between two adjacent battery modules **3**, and the separator **12** is attached to a side surface of the battery module **3**. The separator **12** is provided so that the separator **12** may separate the space within the box body **1**, and each battery module **3** is separately restrained, thereby avoiding the direct contact between the battery modules **3**, improving the restraining effect on the battery module **3**, and thus improving the reliability of the battery pack.

[0078] In this embodiment, the box body **1** is a cuboid, the box body **1** is formed by splicing and

enclosing six splice plates **11**, and each separator **12** is fixed and connected to a corresponding splice plate **11**.

Claims

- 1.** A battery module, comprising a plurality of battery cells, a plurality of first connection members and a plurality of second connection members; wherein a battery cell of the plurality of battery cells has an end surface, and the end surface is provided with a tab; the plurality of first connection members and the plurality of second connection members are flexible members, a first connection member of the plurality of first connection members comprises a first connection portion, a second connection portion and a bend, the first connection portion and the second connection portion are in sheet-like form, and two ends of the bend are connected to the first connection portion and the second connection portion, respectively; and the plurality of battery cells are disposed in a superimposed manner, the first connection portion is connected to tabs of two adjacent battery cells among the plurality of battery cells, the plurality of second connection members are in sheet-like form, a second connection member of the plurality of second connection members is configured to connect second connection portions of two adjacent first connection members among the plurality of first connection members, and the second connection member, the first connection portion and the second connection portion are superimposed on the end surface of a respective battery cell.
- 2.** The battery module of claim 1, further comprising a supporting member, wherein the battery cell comprises a body portion and an edge sealing portion, a thickness of the body portion is greater than a thickness of the edge sealing portion, the supporting member is disposed between edge sealing portions of two adjacent battery cells among the plurality of battery cells, and a side surface of the supporting member is attached to at least one of a respective first connection member or a respective second connection member.
- 3.** The battery module of claim 2, wherein the supporting member is a flexible supporting block.
- 4.** The battery module of claim 1, wherein a first buffer member is disposed between two adjacent battery cells among the plurality of battery cells, and the first buffer member is in sheet-like form.
- 5.** The battery module of claim 1, wherein the tab comprises a third connection portion and a fourth connection portion, the fourth connection portion is connected to a core within the battery cell through the third connection portion, the third connection portion and the fourth connection portion are in sheet-like form, the fourth connection portion is perpendicularly connected to the third connection portion, the fourth connection portion is parallel to the end surface of the battery cell, and the fourth connection portion is connected to the first connection portion.
- 6.** A battery pack, comprising a box body and a battery module, wherein the battery module is disposed in the box body, and a side surface of the battery module is attached to an inner wall of the box body; wherein the battery module comprises a plurality of battery cells, a plurality of first connection members and a plurality of second connection members; wherein a battery cell of the plurality of battery cells has an end surface, and the end surface is provided with a tab; the plurality of first connection members and the plurality of second connection members are flexible members, a first connection member of the plurality of first connection members comprises a first connection portion, a second connection portion and a bend, the first connection portion and the second connection portion are in sheet-like form, and two ends of the bend are connected to the first connection portion and the second connection portion, respectively; and the plurality of battery cells are disposed in a superimposed manner, the first connection portion is connected to tabs of two adjacent battery cells among the plurality of battery cells, the plurality of second connection members are in sheet-like form, a second connection member of the plurality of second connection members is configured to connect second connection portions of two adjacent first connection members among the plurality of first connection members, and the second connection member, the first connection portion and the second connection portion are superimposed on the end surface of

a respective battery cell.

7. The battery pack of claim 6, further comprising a wire and a second buffer member, wherein a current output seat is disposed on the box body, the battery module has a power transmission side surface, and the plurality of second connection members of the battery module are disposed on the power transmission side surface, two ends of the wire are connected to the current output seat and the plurality of second connection members, respectively, and the second buffer member is disposed between the power transmission side surface and the inner wall of the box body.

8. The battery pack of claim 7, wherein a wiring groove is disposed within the second buffer member, and the wire is arranged within the wiring groove.

9. The battery pack of claim 7, wherein the second buffer member is provided with a weight reduction hole.

10. The battery pack of claim 6, wherein a plurality of battery modules are provided, a plurality of separators are also disposed within the box body, the plurality of separators are fixed to the inner wall of the box body, a separator of the plurality of separators is disposed between two adjacent battery modules, and the separator is attached to the side surface of a respective battery module of the plurality of battery modules.

11. The battery pack of claim 6, further comprising a supporting member, wherein the battery cell comprises a body portion and an edge sealing portion, a thickness of the body portion is greater than a thickness of the edge sealing portion, the supporting member is disposed between edge sealing portions of two adjacent battery cells among the plurality of battery cells, and a side surface of the supporting member is attached to at least one of a respective first connection member or a respective second connection member.

12. The battery pack of claim 11, wherein the supporting member is a flexible supporting block.

13. The battery pack of claim 6, wherein a first buffer member is disposed between two adjacent battery cells among the plurality of battery cells, and the first buffer member is in sheet-like form.

14. The battery pack of claim 6, wherein the tab comprises a third connection portion and a fourth connection portion, the fourth connection portion is connected to a core within the battery cell through the third connection portion, the third connection portion and the fourth connection portion are in sheet-like form, the fourth connection portion is perpendicularly connected to the third connection portion, the fourth connection portion is parallel to the end surface of the battery cell, and the fourth connection portion is connected to the first connection portion.
